Z-J-191D
October 26, 1978
SUPERSEDING
Fed. Spec. Z-J-191C
October 7, 1974

FEDERAL SPECIFICATION

JELLIES, FRUIT

This specification was approved by the Commissioner, Federal Supply Service, General Services Administration, for the use of all Federal agencies.

1. SCOPE AND CLASSIFICATION

1.1 Scope. This specification covers the requirements for fruit jellies for use by all Federal agencies.

1.2 Classification. Fruit jellies shall be of the following types, kinds and grades, as specified (see 6.1):

1.2.1 Types. The jelly shall be of the following types:

Type I - Prepared from a single kind of fruit
Type II - Prepared from a mixture of two or more kinds of fruit

1.2.2 Kinds. The jelly shall be of the following kinds:

(1) Apple
(2) Blackberry
(3) Boysenberry
(4) Cherry
(5) Crabapple
(6) Currant, red
(7) Dewberry
(8) Gooseberry
(9) Grape (Concord)
(10) Grape (other than Concord)
(11) Guava
(12) Loganberry
(13) Pineapple
(14) Plum (Damson)
(15) Plum (Greengage)
(16) Plum (other than Damson, Greengage or Prune)

FSC 8930

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(17) Quince
(18) Raspberry, black
(19) Raspberry, red
(20) Raspberry (of reddish-purple varieties)
(21) Strawberry
(22) Others not listed

1.2.3 Grades. The jellies shall be of the following grades:

Grade A
Grade B

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein:

Federal Specifications:

Z-J-96	-	Jams (Preserves); Fruit
PPP-B-636	-	Boxes, Shipping, Fiberboard
PPP-C-29	-	Canned Subsistence Items, Packaging and Packing Of
PPP-G-460	-	Glass Containers, Filled and Closed, Packaging and Packing Of
PPP-I-350	-	Individual Servings of Subsistence; Packaging and Packing Of

Federal Standard:

FED-STD-123	-	Marking for Shipment (Civil agencies)

(Activities outside the Federal Government may obtain copies of Federal Specifications, Standards, and Handbooks as outlined under General Information in the Index of Federal Specifications and Standards and at the prices indicated in the Index. The Index, which includes cumulative monthly supplements as issued, is for sale on a subscription basis by the Superintendent of Documents, US Government Printing Office, Washington, DC 20402.

(Single copies of this specification and other Federal specifications required by activities outside the Federal Government for bidding purposes are available without charge from Business Service Centers at the General Services Administration Regional Offices in Boston, New York, Philadelphia, Washington, DC, Atlanta, Chicago, Kansas City, MO, Forth Worth, Houston, Denver, San Francisco, Los Angeles, and Seattle, WA.

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Military Specifications:

MIL-L-1497 - Labeling of Metal Cans for Subsistence Items
MIL-L-35078 - Loads, Unit: Preparation of Nonperishable Subsistence

Military Standards:

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129 - Marking for Shipment and Storage
MIL-STD-668 - Sanitary Standards for Food Plants

(Copies of Military Specifications, Standards, and publications required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

LAWS AND REGULATIONS

US Department of Health, Education and Welfare

Federal Food, Drug and Cosmetic Act and Regulations Promulgated Thereunder

21 CFR 150.140 - Standards of Identity - Fruit Jelly
21 CFR 110 - Current Good Manufacturing Practice (Sanitation) in Manufacturing, Processing, Packing, or Holding Human Food

(The Code of Federal Regulations (CFR) and the Federal Register (FR) are for sale on a subscription basis by the Superintendent of Documents, U.S. Government Printing Office, Washington, DC 20402. When indicated, reprints of certain regulations may be obtained from the Federal agency responsible for issuance thereof.)

US Department of Agriculture

United Stated Standards for Condition of Food Containers
Regulations Governing Inspection and Certification of Processed Fruits and Vegetables and Related Products
United States Standards for Grades of Fruit Jelly

(Application for copies should be addressed to the Information Division, Food Safety and Quality Service, US Department of Agriculture, Washington, DC 20250.)

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2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless a specific issue is identified, the issue in effect on date of invitation for bids or request for proposal shall apply:

National Motor Freight Traffic Association, Inc., Agent

National Motor Freight Classification

(Application for copies should be addressed to the American Trucking Associations, Inc., Traffic Department, 1616 P Street, N.W., Washington, DC 20036.)

Uniform Classification Committee, Agent

Uniform Freight Classification

(Application for copies should be addressed to the Uniform Classification Committee, Room 1106, 222 South Riverside Plaza, Chicago, IL 60606.)

Association of Official Analytical Chemists

Official Methods of Analysis of the Association of Official Analytical Chemists

(Application for copies should be addressed to the Association of Official Analytical Chemists, PO Box 540, Benjamin Franklin Station, Washington, DC 20044.)

3. REQUIREMENTS

3.1 Materials. All materials shall be clean, sound and wholesome, and free from foreign odors, foreign flavors, insect infestation or objectionable foreign matter. In addition, the following special quality factors are listed below:

3.1.1 Fruit juices. The fruit juice shall be prepared from clean, sound and mature, properly prepared, fresh, frozen or canned fruits. The juice shall be the filtered or strained liquid, which may have been depectinized and which have been extracted with or without the application of heat and with or without the addition of water. Concentrated juices may also be used providing they have been properly prepared and are used in proper proportions. Any grape juice used for jelly shall have been treated for precipitation of potassium bitartrate.

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3.1.1.1 Type II jelly. Type II jelly shall be prepared from two or more kinds of fruit juice with the weight of each not less than 1/5 the weight of the combination except that when apple juice is used, its weight shall not be more than one-half the weight of the fruit juice combination.

3.1.2 Sweetening ingredients. Sweetening ingredients may be any of those permitted, in the quantities permitted by the Food and Drug Standards of Identity for Fruit Jelly unless sugar (sucrose) is specifically required by the invitation for bids.

3.1.3 Optional ingredients. Any of the optional ingredients permitted by the Food and Drug Standards of Identity for Fruit Jelly may be used except that sodium potassium tartrate shall not be used and chemical preservatives may be added only to individual serving packages.

3.2 Finished product requirements. Unless otherwise specified, the interpretation and determination of requirements, including descriptive terms and recommendations, shall be in accordance with the United States Standards for Grades of Fruit Jelly, except as modified herein, and lot inspection and certification tolerances applicable thereto, as set forth in the Regulations Governing Inspection and Certification of Processes Fruits and Vegetables and Related Products. The soluble solids of the fruit jelly shall not be less than 65 percent, when determined in accordance with the refractometric method.

3.2.1 Chemical requirements (Military agencies). The jelly, on a lot average basis, shall meet the following minimum requirements:

TABLE I. Minimum chemical requirements (Military agencies)

Kind of fruit 1/ 2/	Ash percent	K_2O mg per 100 grams	P_2O_5 mg per 100 grams
Apple	0.109	59.0	6.4
Blackberry	0.126	60.0	11.2
Crabapple	0.143	82.0	9.0
Currant, red	0.174	91.0	7.9
Grape (Concord)	0.088	45.0	7.8
Guava	0.213	84.0	7.6
Loganberry	0.165	78.0	8.8
Pineapple	0.141	69.0	7.0
Plum, damson	0.206	112.0	9.0
Quince	0.179	91.0	11.1
Raspberry, black	0.156	90.0	9.4
Raspberry, red	0.145	75.0	15.9
Strawberry	0.158	82.0	9.8

1/ Chemical requirements are applicable only to the listed kinds of jelly.

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2/ Type II jelly shall contain the above constituents in amounts proportionate to the fruit juice in the mixture. Chemical requirements shall be disregarded if one or more of the fruit juices is not listed above.

3.3 Federal Food, Drug and Cosmetic Act. All deliveries shall conform in every respect to the provisions of the Federal Food, Drug and Cosmetic Act and regulations promulgated thereunder. The product shall be prepared in accordance with 21 CFR 110 Current Food Manufacturing Practice (Sanitation) in Manufacturing, Processing, Packing or Holding Human Food.

3.4 Workmanship. For Military agencies, the product shall be processed in establishments meeting the requirements of MIL-STD-668.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Unless otherwise specified (see 6.1), the contractor is responsible to furnish an official inspection certificate issued by the US Department of Agriculture certifying that the grade of the product supplied is as required by the contract or order.

4.2 Preacceptance inspection (plant sanitation inspection).

4.2.1 Civil agencies. When specified (see 6.1), inspection of plants producing the product will be conducted upon request of the contracting officer to the appropriate Government inspection agency. Award of contract will be limited to establishments known to maintain proper sanitary conditions or which have received prior sanitary approval recommended by the inspection agency.

4.2.2 Military agencies. The product shall be rejected if produced in plants not meeting the requirements of 3.4.

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4.3 Acceptance inspection.

4.3.1 Condition inspection of containers. When inspection of product for quality and condition is performed by the US Department of Agriculture, the United States Standards for Condition of Food Containers shall apply. (For applicable criteria governing inspection of containers for Military agencies, see 4.3.4).

4.3.2 Sampling for inspection. Sampling for inspection shall be performed in accordance with MIL-STD-105, except where otherwise indicated hereinafter.

4.3.3 Component and material inspection. In accordance with 4.1, components and materials shall be inspected in accordance with the requirements of referenced specifications, drawings and standards unless otherwise excluded, amended, modified or qualified in this specification or applicable purchase document.

4.3.3.1 Ingredients. Conformance of ingredients (juices, sweetening, optional) to requirements as concerns kinds, composition and quality, as applicable, shall be ascertained by examination of labels, invoices and ingredients etc., as necessary. Use of one or more ingredients determined to be in nonconformance with pertinent requirements shall be cause for rejection of the involved finished product.

4.3.3.2 Formulation. Examination shall be performed to the extent necessary to determine conformance to formulation requirements (percentage of each kind of juice for type II product, ratio of juices to sweetening ingredients and kinds and amounts, as applicable, of optional ingredients). Formulation records shall be maintained. Nonconformance shall be cause for rejection of the involved finished product.

4.3.4 Examination of the end item. Examination of the end item shall be in accordance with table III and 4.3.4.1 through 4.3.4.6. Sampling plans for table III shall be in accordance with table II. The acceptable quality levels (AQLs) shall be expressed as percent defective for table III. A lot shall consist of not more than one kind of jelly, unless an assortment is specified (see 6.1).

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TABLE II. Sampling plan

Table	Sample unit	Lot size expressed in	Inspection level	AQLs Major	AQLs Minor
III	One filled can or glass container	Cans or glass container	S-3	--	2.5

TABLE III. Examination for net weight of can or glass container 1/

Category	Defect
Minor	
201	More than 1/4 ounce (7 g) under specified net weight

1/ Results shall be reported to the nearest 1/4 ounce (7 g).

Lot shall be rejected if sample data indicate a lot average net weight less than specified net weight.

4.3.4.1 Examination for product grade. The examination for grade shall be in accordance with the United States Standards for Grades of Fruit Jelly. A grade lower than that specified shall be cause for rejection of the lot.

4.3.4.2 Examination of primary container. The examination of primary containers for condition and labeling shall be in accordance with the United States Standards for Condition of Food Containers. Examination of cans for vacuum and interior enamel shall be in accordance with the applicable quality assurance provisions of PPP-C-29.

4.3.4.3 Examination of preservation and packing for individual servings. Examination shall be in accordance with the end item examination criteria in PPP-I-350, except that for level C pack, only the examination criteria for net weight and shipping containers shall apply. When an assorted pack has been specified, examination shall be conducted to insure that apple and mixed fruit units do not exceed 40 percent of the total. Lot size shall be expressed in intermediate containers or shipping containers, if one and the same, with an inspection level of S-2. The lot shall be rejected if sample data average indicates that the combination of apple and mixed fruit units exceed 40 percent of the total.

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4.3.4.4 Examination for packing and shipping containers for cans. Examinations shall be in accordance with the applicable quality assurance provisions in PPP-C-29.

4.3.4.5 Examination for packing and shipping containers for glass containers. Examinations shall be in accordance with the quality assurance provisions for the end item, as contained in PPP-G-460.

4.3.4.6 Examination of unit loads. Unit loads shall be examined in accordance with the quality assurance provisions of MIL-L-35078.

4.3.5 Testing of finished product. The finished product shall be tested for soluble solids and those finished products listed in table I shall be tested for ash, potassium oxide and phosphorus pentoxide in accordance with 4.4. Test requirements shall be lot average requirements. The sample for testing shall be a one-pound (454 g) composite, derived from the number of primary containers indicated by inspection level S-2 (S-3 for individual servings). Lot size shall be expressed in terms of primary containers. Test results shall be reported to the same unit or decimal place as specified for the pertinent requirement. A test failure shall be cause for rejection of the lot.

4.4 Tests. Test procedures and controls which differ from those specified herein may be used by the supplier if they provide a quality assurance equivalent to that specified. If the Government contracting officer determines that such procedures and controls do not provide, as a minimum, such quality assurance, the supplier will use the test procedures set forth herein. In case of dispute as to test results, the test methods specified herein will govern.

4.4.1 Chemical and physical analyses. Chemical and physical analyses, if required by the purchaser, shall be made in accordance with the methods outlined in the Official Methods of Analysis of the Association of Official Analytical Chemists, as follows:

Test	Source		Method
Ash, percent	Chapter: Section:	Fruit and Fruit Products Ash	
Potassium oxide (K_2O) mg/100 g	Chapter: Section:	Fruit and Fruit Products Potassium	Rapid Flame Photometric

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Test	Source		Method
Phosphorus Pentoxide (P_2O_5) mg/100 g	Chapter: Section:	Fruit and Fruit Products Phosphorus	Volumetric
Soluble Solids, percent	Chapter: Section:	Fruit and Fruit Products Soluble Solids	Refractometer

5. PACKAGING

5.1 Preservation. The jelly, of the flavor specified, shall be preserved in acccordance with 5.1.1, 5.1.2, or 5.1.3, as specified (see 6.1). Applicable to metal and glass containers, a minus one-quarter ounce (7 g) tolerance will be allowed in any one container, provided the average net weight of the containers inspected in accordance with table III is not less than the net weight specified.

5.1.1 Individual servings, levels B and C. Individual servings of jelly shall be preserved in accordance with the level B or C requirements of PPP-I-350, as specified (see 6.1).

5.1.2 Cans, levels A and C. Unless otherwise specified (see 6.1), the jelly, in the quantity and can sizes specified, shall be preserved in accordance with the level A or C requirements of PPP-C-29.

5.1.3 Glass containers, level C. Two pounds (907.2 g) or ten ounces (283.5 g) of jelly as specified (see 6.1), shall be preserved in a glass jar in a manner to protect it from deterioration and damage during shipment from the supply source to the first consuming activity. The supplier may use his standard practice when it meets this requirement.

5.2 Packing. Jelly, of the flavor specified, shall be packed in accordance with 5.2.1, 5.2.2, or 5.2.3, as applicable.

5.2.1 Individual servings, levels B and C. Individual servings of the jelly shall be packed in accordance with the level B or C requirements of PPP-I-350, as specified (see 6.1). When specified (see 6.1), assorted types and kinds of jellies may be packed in the same shipping container. Individual servings of fruit jams conforming to Z-J-96 may be included in the assortment. However, not more than 40 percent of the assorted pack may be apple or mixed fruit units or a combination of the two.

5.2.1.1 When specified (see 6.1 and 6.3), the shipping container specified in 5.2.1 shall be reinforced in accordance with the appendix of PPP-B-636 with nonmetallic strapping or pressure-sensitive, adhesive-filament, reinforced tape except that V3c, V3s or V4s boxes may be reinforced with two bands, one lengthwise and one girthwise.

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5.2.2 Cans, levels A, B, and C. Cans of jelly shall be packed in accordance with the level A, B, or C requirements of PPP-C-29, as specified (see 6.1).

5.2.3 Glass jars, levels A, B and C. Glass jars of jelly shall be packed in accordance with the levels A, B or C requirements of PPP-G-460, as specified (see 6.1).

5.3 Unit loads. When specified (see 6.1), the jelly, packed as specified in 5.2, shall be arranged in unit loads in accordance with MIL-L-35078 for the type and class of load specified.

5.4 Labeling and marking.

5.4.1 Individual servings. Labeling that complies with the Federal Food, Drug and Cosmetic Act and regulations promulgated thereunder is acceptable.

5.4.2 Cans. For military activities, cans shall be labeled in accordance with MIL-L-1497. For civil agencies, labeling shall comply with the Federal Food, Drug and Cosmetic Act and regulations promulgated thereunder.

5.4.3 Glass jars. Labeling that complies with the Federal Food, Drug and Cosmetic Act and regulations promulgated thereunder is acceptable.

5.4.4 Shipping containers. Shipping containers shall be marked in accordance with MIL-STD-129 or FED-STD-123, as applicable. Any additional marking shall be in accordance with PPP-I-350 or PPP-G-460, as applicable.

5.4.5 Unit loads. Unit loads shall be marked in accordance with MIL-L-35078.

6. NOTES

6.1 Ordering data. Procurement documents should specify the following:

(a) Title, number, and date of this specification.
(b) Type, kind, and grade of product desired (see 1.2.1, 1.2.2 and 1.2.3).
(c) When official inspection certificate (USDA) is not required (see 4.1.1).
(d) When final inspection shall be made prior to delivery to point of destination.
(e) When plant sanitation inspection shall be performed by Civil agencies (see 4.2.1).
(f) Type of packaging required (see 5.1).
(g) Levels of preservation and packing required (see 5.1 and 5.2).

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(h) When two pound (907.2 g) or 10 ounce (283.5 g) quantities are required (see 5.1.3).
(i) Type and class of unit load when unit loading is specified (see 5.3).
(j) When an assorted pack is desired (see 5.2.1).

6.1.1 Artificially sweetened (dietetic) fruit jelly. Artificially sweetened fruit jellies are not covered by this specification. Such items, when needed, may be purchased from qualified suppliers on the basis of special applicable requirements.

6.2 Destination inspection (Civil agencies only). When the finished product has been inspected and passed at a point other than destination, the contract should require that the product be inspected at destination for condition and quantity only.

6.3 Military agencies. Based on conditions known or expected to be encountered during shipment, handling, and storage of the specific item being procured, the contracting officer should select the appropriate level of pack in accordance with the criteria established in AR 700-15/NAVSUPINST 4030.28/AFR 71-6/MCO 4030.14D/DSAR 4145.7.

6.4 Packing specified in 5.2.1.1 is intended for transfer at sea operations or specific overseas operation.

Custodians:

Army - GL
Navy - SA
Air Force - 50

Review activities:

Army - MD
Navy - MC, MS
DP - SS

Preparing activity:

Army - GL

Civil Agency Coordinating Activities:

GSA-FSS
VA-DMS

Project No. 8930-0027

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